

hw5 537

gwf

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These data are from an agricultural experiment. The response is the dry weight of vegetable matter produced. The factors are cultivar (or a variety of plant that can be cultivated), soil inoculant (a substance put in the soil to encourage growth), and field. There are two cultivars (a and b), 3 soil inoculants (con(trol), dea(d), and liv(e)), and four fields. The data are in a file called ‘cultivar.dat’. You can no longer ignore field. The four fields should be treated as a random effect. That is, you want to make inference about the treatment effects to all possible fields, not just the four fields in the experiment.

1. Use lmer to fit a model on the 2×3 experiment with a random effect for field. What is the variance component for random error (the within field error)?
2. What is the between field variance component?
3. Now do a cell means analysis still accounting for the variance due to fields. What are the estimates of the six cell means?
4. Create the $\hat{\Sigma}$ matrix for the data using the estimated variance components. What are the dimensions of the matrix?
5. What is the estimated covariance matrix of the six cell means?
6. Create a contrast matrix to test the effect of cultivars. Is the test significant?
7. Create a contrast matrix to test the effect of inoculents. Is the test significant?
8. Create a contrast matrix for the interactions. (We should have done this first.) Are there significant interactions?
9. Now eliminate all the data in the cultivar b inoculent liv(e) cell. Find the estimates of the variance components for field and error.
10. Find the estimates for the cell means in the remaining 5 cells using the appropriate model.

11. Now you need to set one of the degrees of freedom for interaction in the full model analysis to 0. Under that constraint estimate the mean of the cultivar b inoculent liv(e) cell. What is the estimate for that cell? How does it compare to the cell estimate when all the data are present?
12. What is the estimated standard error associated with that cell mean? What is the estimated standard error associated with the cultivar a inoculent liv(e) cell?